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guidelines summaries

Biomarkers for Adjuvant Endocrine and Chemotherapy in Early-Stage Breast Cancer: ASCO Guideline Update Q and A

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Since the 2016 guidelines on biomarkers for patients with early-stage breast cancer, new data have been reported that offer additional guidance on genomic testing (Onco*type* Dx 21-gene Recurrence Score [RS], MammaPrint, Prosigna, EndoPredict, and Breast Cancer Index [BCI]) to inform adjuvant endocrine and chemotherapy use, on the basis of patient age and number of involved axillary lymph nodes.[1] The goal of the current guidelines is to update endocrine and chemotherapy recommendations in individuals with estrogen receptor (ER)–positive, human epidermal growth factor receptor 2 (HER2)–negative tumors, and in those with HER2-positive or triple-negative breast cancer.[2] The questions below will focus on the management of ER-positive breast cancer.

QUESTION: FOR PATIENTS WHO ARE PREMENOPAUSAL OR AGE ≤ 50 YEARS WITH NEWLY DIAGNOSED ER-POSITIVE, HER2-NEGATIVE, LYMPH NODE–NEGATIVE BREAST CANCER, WHICH BIOMARKERS SHOULD BE USED TO GUIDE DECISIONS ON ADJUVANT ENDOCRINE AND CHEMOTHERAPY?

In TAILORx, participants age ≤ 50 years with ER-positive, HER2-negative, lymph node–negative breast cancer had detectable chemotherapy benefit if the RS was from 16 to 25.[3] Participants in TAILORx with RS 26-100 were treated with chemoendocrine therapy and had superior outcomes than anticipated with endocrine therapy alone.[4] Incorporating clinical risk offered additional prognostic information to RS and further informed the absolute chemotherapy benefit to women age ≤ 50 years and RS 11-25.[5] RSClin integrates RS 0-100 with tumor grade, tumor size, and age to individualize risk and guide discussions regarding adjuvant chemotherapy benefit for those with lymph node–negative breast cancer.[6]

In an update of MINDACT in women with ER-positive, HER2-negative early-stage breast cancer and 0-3 axillary nodes involved with high clinical, low genomic risk, there was an adjuvant chemotherapy benefit in women age ≤ 50 years but not to those > 50 years.[7] The Panel does not recommend MammaPrint use in this population age ≤ 50 years or in patients with low clinical risk (Fig 1). There are no data from prospective randomized trials to recommend EndoPredict or Prosigna in this population.

QUESTION: FOR PATIENTS WHO ARE PREMENOPAUSAL OR AGE ≤ 50 YEARS WITH NEWLY DIAGNOSED ER-POSITIVE, HER2-NEGATIVE, LYMPH NODE–POSITIVE BREAST CANCER, WHICH BIOMARKERS SHOULD BE USED TO GUIDE DECISIONS ON ADJUVANT ENDOCRINE AND CHEMOTHERAPY?

In the 33% of participants in the RxPONDER trial who were premenopausal, there was an improvement in invasive and distant disease-free survival with the addition of chemotherapy to those with a RS 0-25 and 1-3 involved lymph nodes.[8] Although relative chemotherapy benefit did not increase with higher RS, there was greater absolute chemotherapy benefit observed with higher RS in premenopausal women with RS 0-25. RxPONDER, TAILORx, and MINDACT were not designed to test whether chemotherapy can be replaced by ovarian function suppression in premenopausal women, and the rate of ovarian function suppression was limited.

Although genomic assays may offer prognostic information and be used for shared patient-physician treatment decision making, the Panel consensus is that there is insufficient evidence to recommend a biomarker for use in this population. There are no randomized trials that have reported data evaluating the clinical utility of any genomic assay in patients with ≥ 4 lymph nodes.

QUESTION: FOR PATIENTS WHO ARE POSTMENOPAUSAL OR AGE > 50 YEARS WITH NEWLY DIAGNOSED ER-POSITIVE, HER2-NEGATIVE, LYMPH NODE–NEGATIVE BREAST CANCER, WHICH BIOMARKERS SHOULD BE USED TO GUIDE DECISIONS ON ADJUVANT ENDOCRINE AND CHEMOTHERAPY?

In TAILORx, there was no chemotherapy benefit for postmenopausal women with RS < 26.[3] In MINDACT, the 8-year distant metastasis free–survival was similar in those who received versus those who did not receive adjuvant chemotherapy in patients > 50 years with ER-positive, HER2-negative early-stage breast cancer

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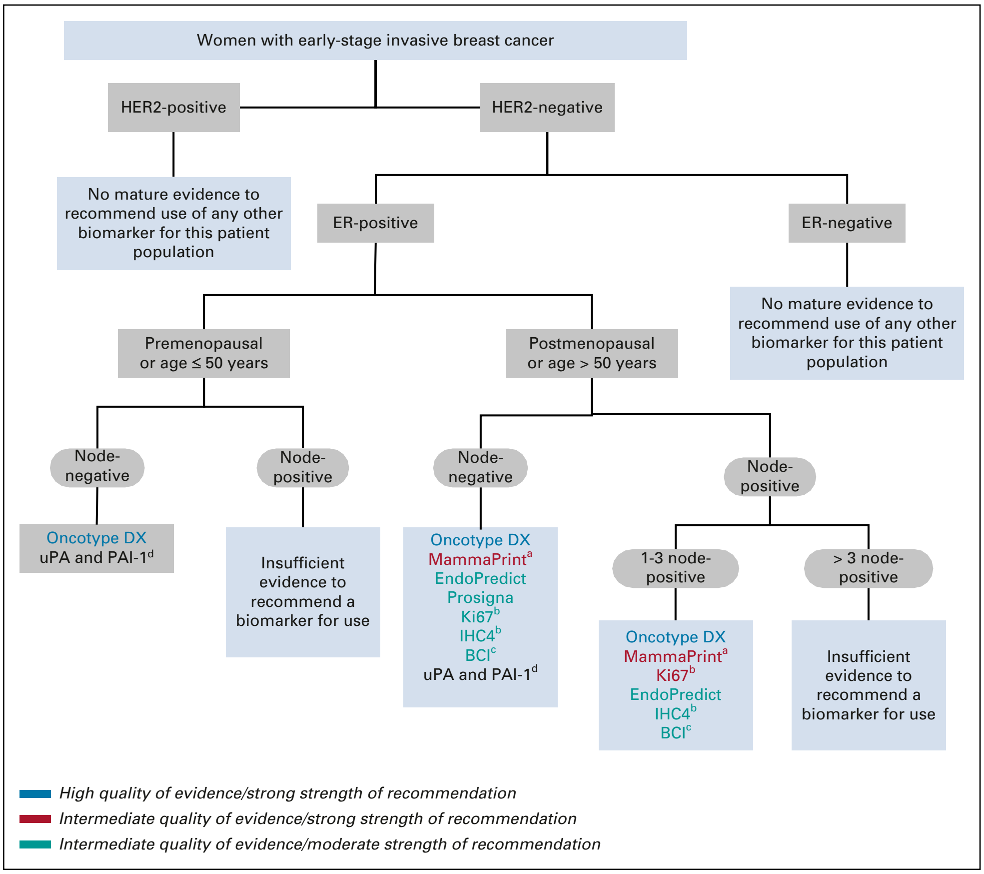

FIG 1. Algorithm on biomarkers to guide decisions on adjuvant endocrine and chemotherapy. The full set of recommendations is available in the Biomarkers for Adjuvant Endocrine and Chemotherapy in Early-Stage Breast Cancer: ASCO Guideline Update.[2] [a]Only in patients with high clinical risk per MINDACT categorization. [b]Only if locally validated and together with other parameters in patients who do not have access to genomic tests. [c]May also be offered to patients who received 5 years of endocrine therapy without evidence of recurrence. [d]This biomarker is no longer in use. BCI, Breast Cancer Index; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; IHC4, immunohistochemistry 4; PAI-1, plasminogen activator inhibitor-1; uPA, urokinase plasminogen activator.

with 0-3 axillary nodes involved and a high clinical risk, low genomic risk.[7] The Panel recommends the use of MammaPrint in this population be limited to those with high clinical risk. Prosigna and EndoPredict demonstrate comparable prognostic capacities, but with less economical and clinical utility studies. No level IA trial data are available yet for these assays compared with MammaPrint and Onco*type*DX.

QUESTION: FOR PATIENTS WHO ARE POSTMENOPAUSAL OR AGE > 50 YEARS WITH NEWLY DIAGNOSED ER-POSITIVE, HER2-NEGATIVE, LYMPH NODE–POSITIVE BREAST CANCER, WHICH BIOMARKERS SHOULD BE USED TO GUIDE DECISIONS ON ADJUVANT ENDOCRINE AND CHEMOTHERAPY?

In the 67% of participants in RxPONDER who were postmenopausal, there was no difference in the 5-year invasive and distant disease-free survival rates for those assigned to chemotherapy or not.[8] For patients with 1-3 positive lymph nodes, the Panel consensus is that there is sufficient evidence for the use of the RS and MammaPrint (in patients with high clinical risk). Prosigna and EndoPredict may also be considered, although evidence quality and strength of recommendation are intermediate and moderate, respectively, because of lack of prospective randomized trials. There is insufficient evidence to recommend any genomic assay in patients with ≥ 4 lymph nodes.

QUESTION: IN PATIENTS WITH ER-POSITIVE, HER2-NEGATIVE BREAST CANCER, WHICH BIOMARKER CAN BE USED TO DETERMINE WHETHER TO EXTEND ENDOCRINE THERAPY?

The benefit from extending adjuvant endocrine therapy is modest and associated with tolerability concerns. The predictive benefit of BCI for extended endocrine therapy has been demonstrated in various cohorts that include patients with 0-3 involved lymph nodes.[9-12] The data to support the utility of BCI in patients with ≥ 4 positive lymph nodes are limited. Similar recommendations for premenopausal women cannot be definitively made, as the majority of patients were postmenopausal in the trials.

QUESTION: SHOULD TESTING FOR Ki67 BE PERFORMED WHEN CONSIDERING ADJUVANT ABEMACICLIB IN A PATIENT WITH ER-POSITIVE, HER2-NEGATIVE, LYMPH NODE–POSITIVE BREAST CANCER?

In 2021, the US Food and Drug Administration (FDA) approved abemaciclib in combination with endocrine therapy for adjuvant treatment in patients with ER-positive, HER2-negative, lymph node–positive breast cancer at high risk of recurrence and Ki67 ≥ 20%, as determined by an FDA-approved test (currently Agilent Technologies [formerly DAKO], MIB-1 anti-Ki67 antibody in a proprietary automated platform). This approval was based on the MonarchE trial, which demonstrated an improvement in invasive and distant disease-free survival rate in this population.[13] Ki-67 was prognostic but not predictive in MonarchE. Broad implementation of the FDA-approved test for Ki67 may be challenging, and Ki-67 testing performed by an experienced pathologist using other laboratory-developed tests may perform equally as well as the FDA-approved test when cross-validated.

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REFERENCES

1. Harris LN, Ismaila N, McShane LM, et al: Use of biomarkers to guide decisions on adjuvant systemic therapy for women with early-stage invasive breast cancer: American Society of Clinical Oncology clinical practice guideline. J Clin Oncol 34:1134-1150, 2016
2. Andre F, Ismaila N, Allison KH, et al: Biomarkers for adjuvant endocrine and chemotherapy in early-stage breast cancer: ASCO guideline update. J Clin Oncol 40:1816-1837, 2022
3. Sparano JA, Gray RJ, Makower DF, et al: Adjuvant chemotherapy guided by a 21-gene expression assay in breast cancer. N Engl J Med 379:111-121, 2018
4. Sparano JA, Gray RJ, Makower DF, et al: Clinical outcomes in early breast cancer with a high 21-gene recurrence score of 26 to 100 assigned to adjuvant chemotherapy plus endocrine therapy: A secondary analysis of the TAILORx randomized clinical trial. JAMA Oncol 6:367-374, 2020
5. Sparano JA, Gray RJ, Ravdin PM, et al: Clinical and genomic risk to guide the use of adjuvant therapy for breast cancer. N Engl J Med 380:2395-2405, 2019
6. Sparano JA, Crager MR, Tang G, et al: Development and validation of a tool integrating the 21-gene recurrence score and clinical-pathological features to individualize prognosis and prediction of chemotherapy benefit in early breast cancer. J Clin Oncol 39:557-564, 2021
7. Piccart M, van 't Veer LJ, Poncet C, et al: 70-gene signature as an aid for treatment decisions in early breast cancer: Updated results of the phase 3 randomised MINDACT trial with an exploratory analysis by age. Lancet Oncol 22:476-488, 2021
8. Kalinsky K, Barlow WE, Gralow JR, et al: 21-gene assay to inform chemotherapy benefit in node-positive breast cancer. N Engl J Med 385:2336-2347, 2021
9. Zhang Y, Schroeder BE, Jerevall PL, et al: A novel Breast Cancer Index for prediction of distant recurrence in HR(+) early-stage breast cancer with one to three positive nodes. Clin Cancer Res 23:7217-7224, 2017
10. Sgroi DC, Sestak I, Cuzick J, et al: Prediction of late distant recurrence in patients with oestrogen-receptor-positive breast cancer: A prospective comparison of the breast-cancer index (BCI) assay, 21-gene recurrence score, and IHC4 in the TransATAC study population. Lancet Oncol 14:1067-1076, 2013
11. Bartlett JMS, Sgroi DC, Treuner K, et al: Breast Cancer Index and prediction of benefit from extended endocrine therapy in breast cancer patients treated in the Adjuvant Tamoxifen-To Offer More? (aTTom) trial. Ann Oncol 30:1776-1783, 2019
12. Blok EJ, Kroep JR, Meershoek-Klein Kranenbarg E, et al: Optimal duration of extended adjuvant endocrine therapy for early breast cancer; Results of the IDEAL trial (BOOG 2006-05). J Natl Cancer Inst 110:40-48, 2018
13. Harbeck N, Rastogi P, Martin M, et al: Adjuvant abemaciclib combined with endocrine therapy for high-risk early breast cancer: Updated efficacy and Ki-67 analysis from the monarchE study. Ann Oncol 32:1571-1581, 2021

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